

INTERVAL ESTIMATION

From point estimate to interval estimate [READING]

μ ▶ **Parameter:** θ . Measurable **population** characteristic (e.g. mean, μ)

$\bar{X}$ Based on n random i.i.d. draws from $X, X_1, X_2, \dots, X_n$:

▶ **Point Estimator:** $\hat{\theta}$. **Statistic**, $\hat{\theta} = f(X_1, \dots, X_n)$, used to **estimate** θ (e.g. $\bar{X}$ is an estimator of μ).

$\bar{x}$ ▶ **Point Estimate:** $\hat{\theta}$. **Realisation of** $\hat{\theta}$ in relation to a sample $\hat{\theta} = f(x_1, \dots, x_n)$.

Problem: $\text{Prob}(\hat{\theta} = \theta)$ is null if $\hat{\theta}$ is a continuous r.v., or it is unknown (θ is not known).

Solution:

▶ **Confidence interval:** $(\hat{\theta}_1, \hat{\theta}_2)$. **Random interval** whose extremes are **two statistics** $\hat{\theta}_1 = f_1(X_1, \dots, X_n)$ and $\hat{\theta}_2 = f_2(X_1, \dots, X_n)$, used to **estimate** θ , built in such a way that $\text{Prob}(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2) = 1 - \alpha$ with $0 \leq \alpha \leq 1$ chosen a priori. This interval is called a **100(1 - α)% confidence interval** and it is clearly random, as its extremes depend on the random variables $X_1, \dots, X_n$.

▶ **Interval estimate:** $(\hat{\vartheta}_1, \hat{\vartheta}_2)$ realisation of the interval given a specific sample $x_1, \dots, x_n$: $\hat{\vartheta}_1 = f_1(x_1, \dots, x_n)$ and $\hat{\vartheta}_2 = f_2(x_1, \dots, x_n)$.

$$\bar{X} \pm ME$$

$$\bar{x} \pm ME$$

From point estimate to interval estimate [READING]

The uncertainty associated with a point estimate is due to the sampling variability, i.e. to the different results that can be obtained based on the different samples that can be drawn from a population.

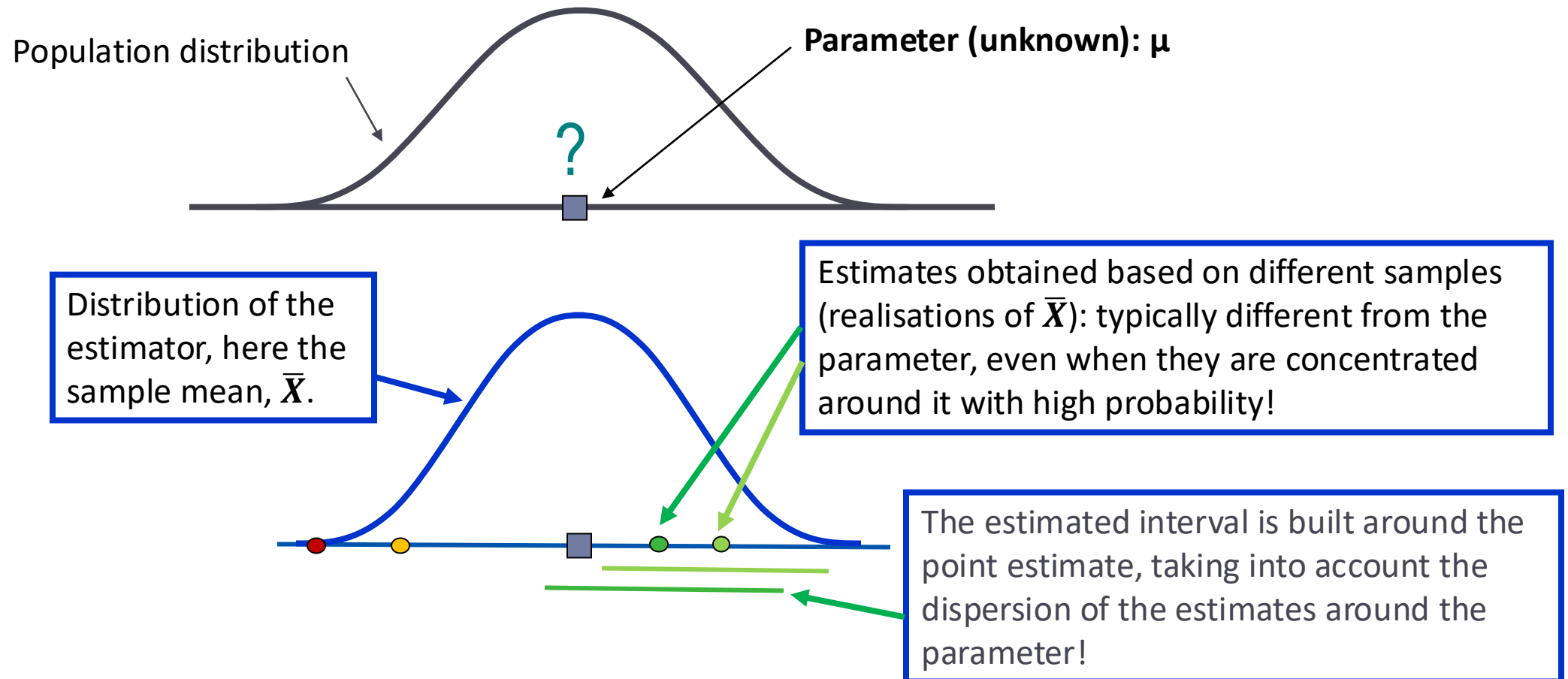
This uncertainty can be quantified by the standard error of the estimator, and also depends on its distribution.

The idea behind confidence intervals is to build an estimator that **explicitly incorporates** this uncertainty.

In many **cases**, and specifically in **all the cases that we will take into account**, the interval estimator can be obtained by adding and subtracting from an (appropriate) **point estimator** an amount called the **margin of error (ME)**, which in turn depends on the **standard error** and on the **distribution of** the estimator

From point estimate to interval estimate [READING]

Consider the case where one wants to estimate the mean, μ , of a **population X**



Confidence interval: general concepts [READING]

Building a confidence interval

Point estimator $\pm$ Margin of error

Point Estimator $\pm$ (Reliability Factor)(Standard Error)

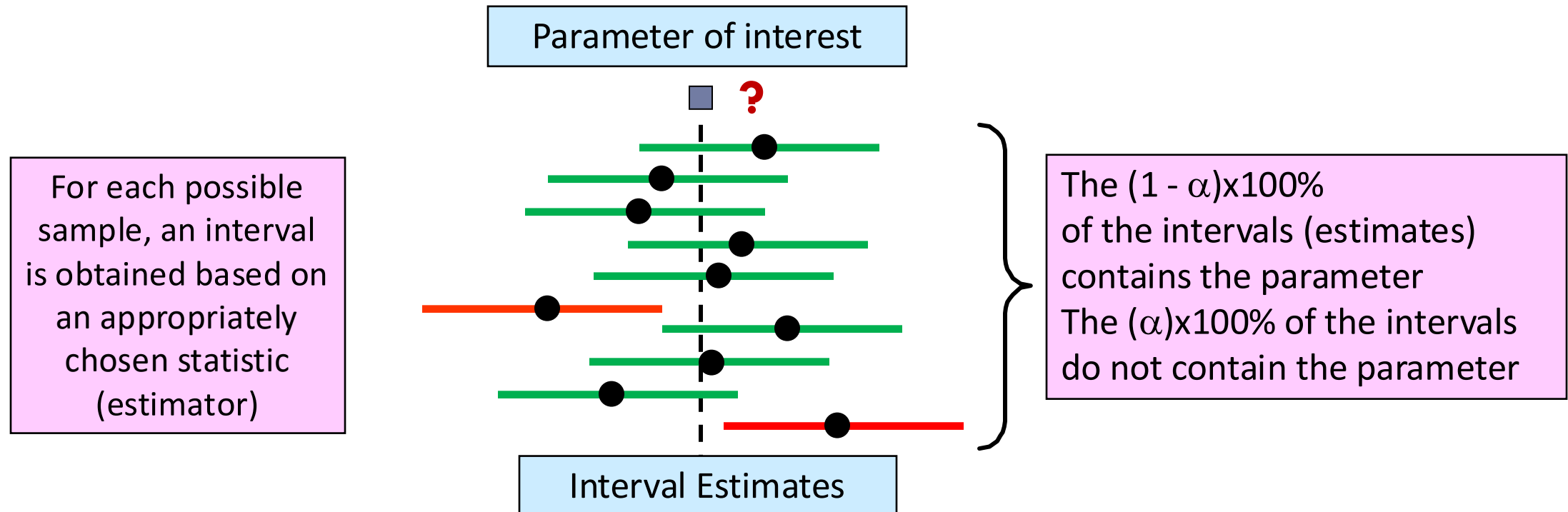
- ▶ The **reliability factor** depends on the **required confidence level**, $(1 - \alpha)$, and on the **distribution of** the point estimator, and – given the standard error – controls (determines) the width of the interval.
- ▶ If the confidence level $(1 - \alpha)$ is high, i.e. if the probability that the interval (estimator) contains the parameter of interest is high, the width of the interval will be larger.
 - ▶ Indeed, the interval $(-\infty ; \infty)$ contains the parameter with probability 1, while a point estimator will cover the parameter with probability 0 (if the point estimator is a continuous r.v.).
 - ▶ There is therefore a **trade-off** between the precision of the interval (the greater the smaller the interval width) and the confidence level.

The confidence level [READING]

The confidence level ($1 - \alpha$)

- ▶ Probability that the **random** confidence interval contains the unknown parameter
- ▶ It can also be seen as the **proportion** of samples (out of all those that can be drawn from the population) leading to interval **estimates** containing the parameter.
- ▶ The interval estimate built based on a **specific** sample may or may not contain the parameter (and it is not possible to state which of the two situations occurs, because the parameter is unknown). Thus, **it makes no sense to associate any probability with the actual interval estimate**

The confidence level: frequentist interpretation [READING]



For a specific interval (calculated from one specific sample) **we cannot know whether or not it contains the parameter. However, if we know that $(1 - \alpha) \times 100\%$ of intervals contain the parameter, we can be *confident at the $(1 - \alpha)$ level* that the observed interval is one of those including the parameter!**

$X_1, X_2, \dots, X_n$ i.i.d. $X_i \sim N(\mu, \sigma^2)$ (or $n \geq 30$, so we apply CLT)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad SE = \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$, a realisation of $\bar{X}$, is our estimate of μ .

Interval estimation

Confidence intervals for the mean [PEPVS 1]

We associate SE to the estimator $\bar{X}$. SE represents the expected distance of the estimator from μ .

IDFA We can combine SE and $\bar{x}$ to provide an interval of values that are plausible values for μ .

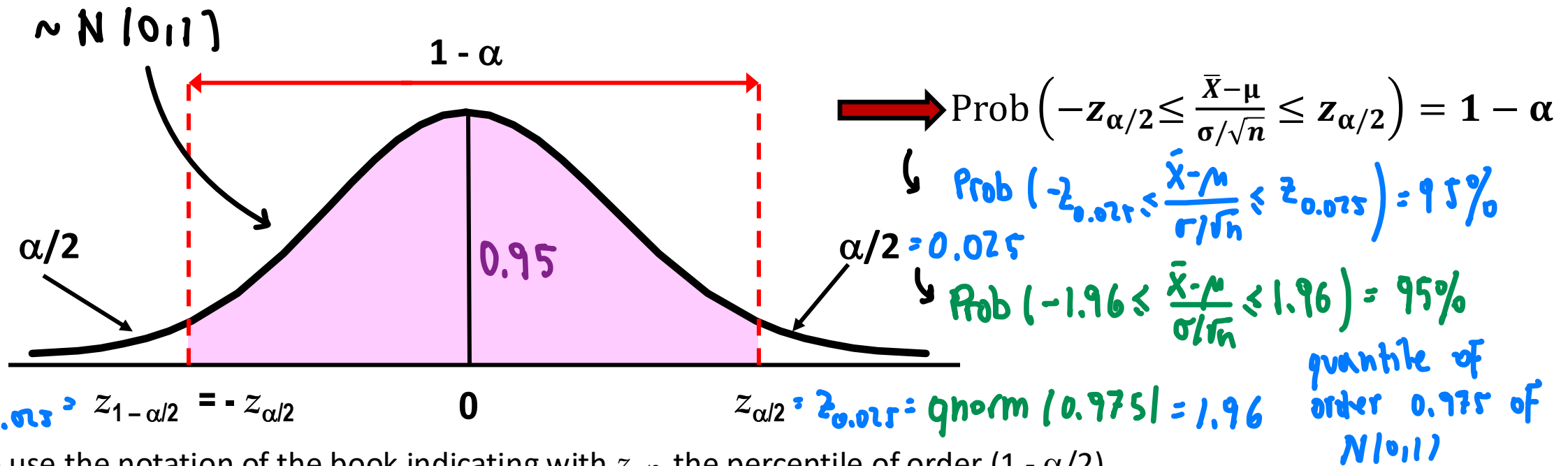
Confidence interval for $\mu : \sigma^2$ known . We consider $\alpha = 0.05$

We want to estimate the mean, μ , of a population X having known variance, σ^2 .

Assume that

- ▶ X has **normal** distribution
- ▶ or the sample is large enough, so that the **central limit theorem** can be applied.

In such cases $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ and consequently $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$



* We use the notation of the book indicating with $z_{\alpha/2}$ the percentile of order $(1 - \alpha/2)$

Confidence interval for $\mu : \sigma^2$ known

$\alpha = \text{SIGNIFICANCE} = 0.05$
 $1 - \alpha = \text{CONFIDENCE} = 0.95$

Interval estimation of the mean, μ , of a population X with known variance, σ^2 .

Case where X is normal or the sample is large enough (central limit theorem).

Given any value of α it is:

$$\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n) \rightarrow \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

$$\rightarrow \text{Prob} \left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96 \right) = 95\%$$

$$\rightarrow \text{Prob} \left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} \right) = 1 - \alpha = 0.95$$

$$\rightarrow \text{Prob} \left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha$$

The interval **estimator** for μ at the level $(1 - \alpha) \times 100\%$ is:

random interval that contains μ
with probability 0.95

CONFIDENCE INTERVAL ESTIMATOR
FOR μ , WITH CONFIDENCE
LEVEL 0.95 ($1 - \alpha$)

$$CI_{1-\alpha}(\mu) = \left[\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = \left[\bar{X} \mp z_{\alpha/2} \cdot SE(\bar{X}) \right]$$

The interval is obtained by subtracting and adding to $\bar{X}$ the **margin of error** $ME = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$,

given by the product between the **reliability factor**, $z_{\alpha/2}$ (depending on the distribution of the (standardised) estimator and on the confidence level), and the **standard error** of $\bar{X}$.

Rstudio: CI.mean()

The function `CI.mean()` in package `UBStats` allows to determine the confidence interval for the expected value of a population:

```
CI.mean(x, sigma=NULL, conf.level=0.95, digits=2, data)
```

where:

- ▶ `x` is a **numeric** vector or the name of one of the columns of the dataframe specified in `data`
- ▶ `sigma` is the value of the standard deviation, **if known**
- ▶ `conf_level` is the confidence level (0.95 by default)
- ▶ `digits` is the number of decimals used to round off the reported statistics

Hands on: estimation of μ with σ^2 known

To assess the efficiency of a call centre, a survey is conducted on a random sample of **300** calls received in one week (dataframe `CallCenter.P` in `Lesson 14-16_Data.Rdata`) for which, among other things, information on the duration (in seconds) of the conversation (`TimeTalk`) is collected. It is **assumed** that the conversation duration has a standard deviation = 150 sec.

Determine the 99% confidence interval for the average conversation duration.

```
library(UBStats)
CI.mean(x=TimeTalk, sigma=150, conf.level = 0.99, data=CallCenter.P)
```

```
> CI.mean(x=TimeTalk, sigma=150, conf.level = 0.99, data=CallCenter.P)
```

Confidence interval for the mean

Confidence level: 0.99

Variance: known

n	xbar	sigma_X	SE	Lower	Upper
300	254.37	150	8.66	232.07	276.68

[Note that the function can also be used to obtain the point estimate and the standard error of the estimator].

The function reports the number of available data, the sample mean, the standard deviation (in this case known), the standard error of the estimator (also obtained in the last lesson) and the lower and upper limits of the interval estimate

Hands on: estimation of μ with σ^2 known

To assess the efficiency of a call centre, a survey is conducted on a random sample of **300** calls received in one week (dataframe `CallCenter.P` in `Lesson 14-16_Data.Rdata`) for which, among other things, information on the duration (in seconds) of the conversation (`TimeTalk`) is collected. The conversation duration is **assumed to** have a standard deviation = 150 sec.

Confidence interval for the mean

Confidence level: 0.99

Variance: known

n	xbar	sigma	SE	Lower	Upper
300	254.37	150	8.66	232.07	276.68

Can we conclude that the average talk time is between 232.07 and 276.68 with probability 0.99?

No, because the observed interval **is not random**, so it either contains μ or it does not. We can, however, say that 99% of the intervals built around the sample mean contain μ ; hence we can quantify the confidence level (or trust) associated with the interval as 99%, defined as the % of interval estimates that contain the parameter.

Hands on: estimation of μ with σ^2 known / CI width

How to obtain a 99% confidence interval for the average conversation time with width 20?

The length of the confidence interval for the mean in the case of known variance is

$$2ME = 2 z_{\alpha/2} \sigma / \sqrt{n}$$

Given the confidence level and the variance of the population, the only quantity on which one can act to reduce the width of the interval is the sample size.

For the width of the interval to be 20, it is necessary that **ME=10**, i.e. that:

$$\frac{z_{\alpha/2} \sigma}{\sqrt{n}} = 10 \rightarrow n \geq \frac{z_{\alpha/2}^2 \sigma^2}{10^2}$$

```
> # sample size needed to obtain ci of width 10
> z_alpha_2 <- qnorm(0.995)
> z_alpha_2
[1] 2.575829
> (z_alpha_2*150/10)^2
[1] 1492.852
```

*A sample of size higher than or equal to **1493** should be drawn.*

Confidence interval for μ : σ^2 known / CI width

Determination of the sample size needed to guarantee a CI at level $1 - \alpha$ with a given margin of error, ME (or, equivalently, with a given width, 2ME).

For the margin of error to be at **most ME** (or, equivalently, for the interval width to be at **most 2ME**), it is necessary to select a sample with size:

$$n \geq \frac{z_{\alpha/2}^2 \sigma^2}{ME^2}$$

Clearly, the result should be rounded up to the next integer!

This result holds even when the distribution is not normal, as **long as the resulting size is big enough to ensure the applicability of the central limit theorem** (otherwise the distribution of the sample mean would not be normal)

Confidence interval for μ (pop. Normal): σ^2 not known

Interval estimation of the mean, μ , of a **normally** distributed population X with variance σ^2 **unknown**

The confidence interval for μ at level $(1 - \alpha) \times 100\%$ is:

$$CI_{1-\alpha}(\mu) = \left[\bar{X} \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

cannot calculate this formula anymore

Since the variance **is unknown**, the extremes of the interval **are not known!!!**

It would seem natural to **substitute to σ its estimate**, based on the sample variance, $S = \sqrt{S^2}$

However, **the distribution of $\bar{X}$ when replacing S to σ is no longer normal (and therefore the percentile used, $z_{\alpha/2}$, is not appropriate here)!**

The following result is valid, yet: if the population has a **normal** distribution, the statistic

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1) \xrightarrow[\text{(increased uncertainty)}]{\text{replace } \sigma \text{ with } S} T = \frac{\bar{X} - \mu}{S / \sqrt{n}} \sim t_{n-1} \quad (\text{no proof of this } \hat{?})$$

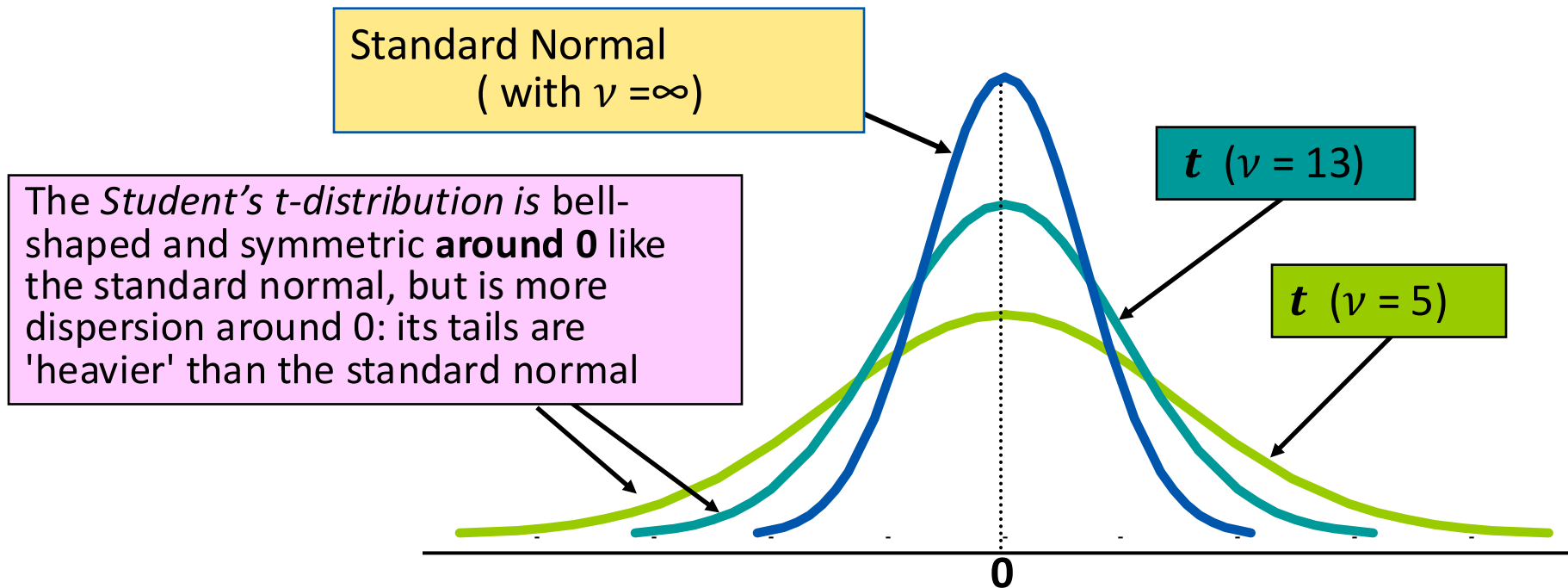
has a known distribution, specifically a **Student's t** distribution with $(n - 1)$ degrees of freedom, denoted by t_{n-1} parameter of student-t

Student's t distribution

Replacing σ^2 with the sample variance, S^2 , increases the uncertainty that characterises $\bar{X}$, since also S^2 varies from sample to sample.

The ***Student's t*** distribution is bell-shaped and centred at **0**, and depends on a single parameter, ν , called **degrees of freedom**, which influences its variability

As the degrees of freedom increase, *Student's t* converges to the standard normal



Confidence interval for μ (pop. Normal): σ^2 not known

Interval estimate for the mean, μ , of a **normally** distributed population with variance, σ^2 , **unknown**

Following the same procedure as in the case where the variance is known and the distribution of the standardised sample mean is a standard normal, when the population distribution is normal and the variance unknown, the confidence interval for μ at the $(1 - \alpha) \times 100\%$ level is:

$$ci_{1-\alpha}(\mu) = \left[\bar{x} \mp t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \right] = \left[\bar{x} \mp t_{n-1, \alpha/2} \cdot se(\bar{X}) \right]$$

The interval is obtained by subtracting and adding to the point estimator, $\bar{X}$, the margin of error $ME = t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$, given by the product between the **reliability factor**, $t_{n-1, \alpha/2}$ – which depends on the distribution of the estimator (standardised using the square root of the sample variance) and on the confidence level – and the **estimate of the standard error** of $\bar{X}$.

Rstudio: CI.mean(), pt(), qt()

The function `CI.mean()` in package `UBStats` allows to determine the confidence interval for the expected value even when the variance is unknown. The syntax is as seen for the case with known variance, except that the argument `sigma` should not be specified.

```
CI.mean(x, conf.level=0.95, digits=2, data)
```

It is also worth introducing the functions available in R to determine the **cumulative probabilities** and the **percentiles** of a Student's t distribution:

`pt(q, df)` and `qt(p, df)`

similar to the functions `pnorm()` and `qnorm()` calculating probabilities and percentiles of a normal distribution, where:

- ▶ **df**: degrees of freedom
- ▶ **q**: (numerical) value at which the cumulative probability has to be calculated. The function `pt()` calculates $F(q) = \text{Prob}(t_{df} \leq q)$ (i.e. the area below the density curve of the Student's t distribution up to q)
- ▶ **p**: order of the **percentile**. The `qt()` function calculates the value at which the cumulative probability is p , i.e. the value $t_{df, 1-p}$ such that $F(t_{df, 1-p}) = \text{Prob}(t_{df} \leq t_{df, 1-p}) = p$.

check
that
the
example

Hands on: estimation of μ (pop. Normal) with σ^2 not known

It is of interest to estimate the average number of customers who visit a shop on weekdays, in order to assess whether it would be worthwhile to hire staff permanently or to increase the number of staff members only at weekends. It is assumed that the number of customers on a weekday has a normal distribution. For a period with no special events, the number of customers on a random sample of weekdays in each week is monitored, for a total of $n = 15$ observations. $\sum x_i = 2755$, $\sum x_i^2 = 585203$ are observed.

95% confidence interval for the average number of customers during a weekday?

```
> n <- 15
> sum_x <- 2755
> sum_x2 <- 585203
> xbar <- sum_x/n
> s2_X <- (sum_x2 - n*xbar^2) / (n-1)

> t_14_alpha_2 <- qt(0.975, df=14)
> ME <- t_14_alpha_2 * sqrt(s2_X/n)
> c((xbar - ME), (xbar + ME))
[1] 142.0142 225.3191
```

$$s^2 = \frac{1}{n-1} \left[\sum x_i^2 - n\bar{x}^2 \right]$$


$$ci_{0.95}(\mu) = \left[\bar{x} \mp t_{14,0.025} \frac{s}{\sqrt{15}} \right]$$


Hands on: estimated μ (pop. Normal) with σ^2 not known

It is of interest to estimate the average number of customers who visit a shop on weekdays, in order to assess whether it would be worthwhile to hire staff permanently or to increase the number of staff members only at weekends. It is assumed that the number of customers on a weekday has a normal distribution. For a period with no special events, the number of customers on a random sample of weekdays in each week is monitored, for a total of $n = 15$ observations. $\sum x_i = 2755$, $\sum x_i^2 = 585203$ are observed.

```
> c ( (xbar-ME) , (xbar+ME) )  
[1] 142.0142 225.3191
```

Is it possible to control the width of the confidence interval acting on n ?

Both $t_{n-1, \alpha/2}$ and s depend on the **extracted sample** and on n . It is therefore not possible to guarantee that an interval will have a specific width, because that will depend on the sample that will be actually extracted and on the resulting variance estimate, which might differ from the actual one. An approximate result can be obtained using the same percentile (as n increases, the percentile decreases) and the current estimate s , which may, however, differ greatly from that estimated from another sample of bigger size.

Confidence interval for μ (non-normal pop.): σ^2 not known

Suppose it is **not** possible to assume that the distribution is normal. Is it possible to obtain confidence intervals for μ in the case when the **variance is unknown**?

- ▶ If n is big enough, for the central limit theorem, $\bar{X}$ has an approximately normal distribution. However, the interval

$$ci_{1-\alpha}(\mu) = [\bar{x} \mp z_{\alpha/2} \sigma / \sqrt{n}]$$

cannot be used because its extremes depend on σ , which is not known

- ▶ However, the sample variance S^2 is an unbiased estimator for σ^2 , and it can be shown that its variance decreases as n increases; thus, when the sample is sufficiently large, variance estimates are concentrated around σ^2 .

In the case of large samples, one can then substitute s to σ to obtain the interval.

Statistical software usually build the interval as in the case of a normal population with σ^2 unknown, thus using the percentile of the Student's distribution, $t_{n-1, \alpha/2}$, which is higher than $z_{\alpha/2}$, in order to take into account the greater uncertainty due to the substitution of σ^2 with its estimate (the two percentiles are however similar when n is high):

$$ci_{1-\alpha}(\mu) = [\bar{x} \mp t_{n-1, \alpha/2} s / \sqrt{n}] \cong [\bar{x} \mp z_{\alpha/2} s / \sqrt{n}]$$

Hands on: estimated μ (pop. not Normal) with σ^2 not known

The dataframe `Insurance` (in `Lesson 14-16_Data.Rdata`) contains information on reimbursements (`refunds`, in dollars) for healthcare costs to (unrelated) subscribers with different socio-demographic characteristics, including the number of `children`.

What is the 95% confidence interval of the average repayment to childless subscribers?

Since no assumptions were made about the distribution of reimbursement in the population, the interval can only be determined if the sample size is sufficiently large

```
> refund.nochild <- Insurance$refund[Insurance$children==0] # data
> length(refund.nochild) # sample width
[1] 574
```

```
> # no need to specify 'date' as x is a vector
```

```
> CI.mean(x=refund.nochild, conf.level=0.95)
```

Confidence interval for the mean

Confidence level: 0.95

Variance: unknown

	n	xbar	s_X	se	Lower	Upper
Normal.Approx	574	12365.98	12023.29	501.84	11382.38	13349.57
Student-t	574	12365.98	12023.29	501.84	11380.30	13351.65

The function reports number of data, sample mean, estimated standard deviation and standard error, and the interval's limits calculated using both normal and Student's t percentiles

Interval estimation

Confidence intervals for the proportion

Confidence interval for the proportion

When one is interested in estimating the proportion, p , of cases with a certain characteristic in the population, the confidence interval can be built based on the sample proportion, $\hat{P}$, i.e. the sample mean of Bernoullian r.v.'s indicating success or failure for the sample units.

When the sample size, n , is sufficiently large, by the central limit theorem:

$$\hat{P} \approx \mathcal{N}(p, p(1-p)/n) \quad \longrightarrow \quad \frac{\hat{P}-p}{\sqrt{p(1-p)/n}} \sim \mathcal{N}(0, 1)$$

The variance of $\hat{P}$ depends on p but, as in the case of a non-normal population with unknown variance, one can estimate the variance by substituting p with the sample proportion, getting:

$$ci_{1-\alpha}(p) = \left[\hat{p} \mp z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] = \left[\hat{p} \mp z_{\alpha/2} \cdot se(\hat{P}) \right]$$

Note that the percentile of the standardised normal distribution is used here.

Rstudio: CI.prop()

The function `CI.prop()` in package `UBStats` allows to determine the confidence interval for the proportion of 'successes' in a population. The syntax is very similar to that of function `CI.mean()`

```
CI.prop(x, success=NULL, conf.level=0.95, digits=2, data)
```

where:

- ▶ **x**: name of a vector or factor or one of the columns of the dataframe specified in **data**
- ▶ **success**: allows you to define which value of **x** is to be considered a success. It can be omitted if **x** is a binary vector (values 0/1; in this case it is assumed that 1 indicates a success) or a logical vector (values **TRUE/FALSE**; in this case it is assumed that **TRUE** indicates a success)
- ▶ **conf_level** is the confidence level (0.95 by default)
- ▶ **digits** is the number of decimals used to round off the reported statistics

Note that the function does not check whether the conditions needed to build the interval (e.g. sample width) are satisfied

Hands on: estimating the proportion

Consider the data on the calls to a call centre in the dataframe `CallCenter.P` (in `Lesson 14-16_Data.Rdata`) with particular reference to the reason for the contact (`Topic`) and to the duration in seconds of the conversation (`TimeTalking`)

Estimate the proportion of telephone calls concerning the contract (`Topic="Contract related"`), p , with a 90% confidence interval.

In order to determine the interval, the sample must be sufficiently large

```
> CI.prop(x=Topic, success="Contract related",  
+         conf.level=0.9,data=CallCenter.P)
```

Confidence interval for the proportion

Confidence level: 0.9

n	pbar	s_X	se	Lower	Upper
300	0.27	0.45	0.03	0.23	0.32

The function reports the sample size (sufficient to allow the application of the central limit theorem), the sample frequency, the estimated standard deviation and standard error, and the limits of the interval

Hands on: estimating the proportion

Consider the data on the calls to a call centre in the `CallCenter.P` dataframe (in **Lesson 14-16_Data.Rdata**) with particular reference to the reason for the contact (**Topic**) and to the duration in seconds of the conversation (**TimeTalking**)

Estimate the proportion of telephone calls lasting more than 6 minutes, p , with a 90% confidence interval.

In order to determine the interval, the sample must be sufficiently large

```
> long.length<-CallCenter.P$TimeTalk>(6*60) # data extraction

> # In this case, it is not necessary to specify the success argument
> # as the vector long.length is logical
> CI.prop(x=long.length,conf.level=0.9)
```

Confidence interval for the proportion
Confidence level: 0.9

n	pbar	s_X	se	Lower	Upper
300	0.27	0.45	0.03	0.23	0.32

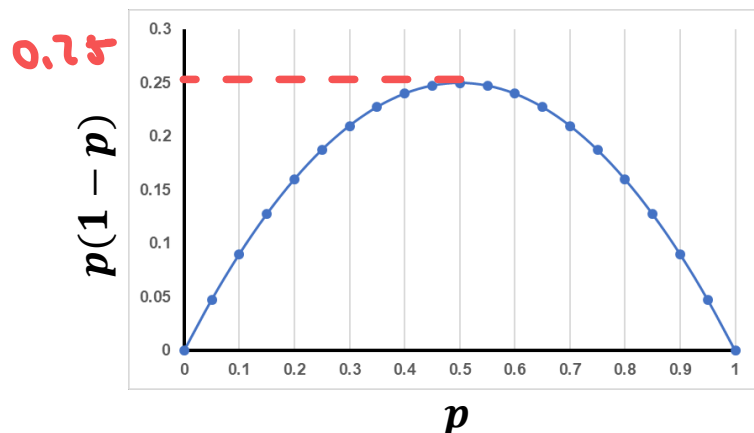
Hands on: proportion estimation / CI width

Is it possible to obtain a 90% confidence interval with width at most 0.05 for the proportion of phone calls lasting more than 6 minutes?

We want to obtain an interval with semi-width $ME=0.025$.

$$ci_{1-\alpha}(p) = \left[\hat{p} \mp z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] \rightarrow ME = z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n} \leq 0.025$$

given the confidence level, **ME** depends on n and also on $\hat{p}$, which depends on the drawn sample.



Consider the function $p(1-p)$.

We note that it reaches its maximum when $p=0.5$. Thus:

$$z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n} \leq z_{\alpha/2} \sqrt{0.5^2/n}$$

Adopting a **conservative** approach, we can determine n by considering the **maximum value of the standard error**

To have $ME \leq 0.025$ ask for
$$h \geq \frac{z_{\alpha/2}^2 0.5^2}{0.025^2}$$

Hands on: proportion estimation / CI amplitude

Is it possible to obtain a 90% confidence interval with width at most 0.05 for the proportion of phone calls lasting more than 6 minutes?

We want to obtain an interval with semi-width $ME=0.025$; using a conservative approach we choose n such that:

$$z_{\alpha/2} \sqrt{\frac{0.5^2}{n}} \leq 0.025 \quad \rightarrow \quad n \geq \frac{z_{\alpha/2}^2 0.5^2}{0.025^2}$$

$0.5^2 \geq \hat{p}(1-\hat{p})$

```
> # sample width to obtain ci of width <=0.05
> z_alpha_2 <- qnorm(0.95)
> z_alpha_2
[1] 1.644854

> (z_alpha_2*0.5/0.025)^2
[1] 1082.217
```

A sample sized 1083 or more guarantees a 90% CI with a width of 0.05, whatever the sample frequency of the extracted sample is (because we obtained n based on the frequency to which the maximum standard error corresponds)

Confidence interval for proportion/ CI width

Determination of the sample size needed to guarantee a CI for the proportion at level $1 - \alpha$ with a given margin of error, ME (or, equivalently, with a given width $2ME$).

Using a conservative approach, for the margin of error to be at **most ME (or, equivalently for the width of the interval to be at most $2ME$)**, it is necessary to select a sample with size:

$$n \geq \frac{z_{\alpha/2}^2 0.5^2}{ME^2}$$

Clearly the result should be rounded up to the next integer!

Interval estimation

Confidence intervals for the difference between means

Confidence interval for the difference between means

As already seen with reference to point estimation, in many applications we are interested in comparing the means of two populations

Population 1 - described by a v.a. X	Population 2 - described by a v.a. Y
$X_1, \dots, X_{n_X}$ sample of size n_X , iid X with mean μ_X and variance σ_X^2	$Y_1, \dots, Y_{n_Y}$ sample of size n_Y iid Y with mean μ_Y and variance σ_Y^2
$\bar{X} = (X_1 + \dots + X_{n_X})/n_X$	$\bar{Y} = (Y_1 + \dots + Y_{n_Y})/n_Y$
$S_X^2 = \sum_{i=1}^{n_X} \frac{(X_i - \bar{X})^2}{n_X - 1}$	$S_Y^2 = \sum_{i=1}^{n_Y} \frac{(Y_i - \bar{Y})^2}{n_Y - 1}$

The parameter of interest is the difference between the means of the two populations, $\mu_X - \mu_Y$
In this case, the confidence interval will be built around the **unbiased** point estimator ($\bar{X} - \bar{Y}$). The reliability factor and the standard error of the estimator depend on **the relationships between the samples and on the assumptions on the distributions of the populations.**

CI for difference between means, independent samples

In the case of **independent** samples, the sample means $\bar{X}$ and $\bar{Y}$ are **independent** and thus **uncorrelated**: the **variance** of the estimator $(\bar{X} - \bar{Y})$ thus results:

$$Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}$$

Moreover, since $\bar{X}$ and $\bar{Y}$ are **independent**, in the case when

- ▶ The two populations X and Y have normal **distribution**
 - ▶ **or** the samples are sufficiently large, so that the **central limit theorem** can be applied.
- one has that:

$$(\bar{X} - \bar{Y}) \sim \mathcal{N}(\mu_X - \mu_Y, [(\sigma_X^2/n_X) + (\sigma_Y^2/n_Y)])$$

Following the same procedure as for building the confidence interval for the mean:

$$\longrightarrow ci_{1-\alpha}(\mu_X - \mu_Y) = \left[(\bar{x} - \bar{y}) \mp z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \right] = \left[(\bar{x} - \bar{y}) \mp z_{\alpha/2} \cdot SE(\bar{X} - \bar{Y}) \right]$$

CI for the difference between means, independent samples

The case when both the variances are known is very **rare**.

- ▶ If the **unknown** variances are assumed to be **equal**, $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, the common variance is estimated using the so-called **pooled corrected sample variance**:

$$s_{Pool}^2 = \frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_X + n_Y - 2} \quad \text{weighted average of sample variances}$$

which is **substituted** to σ_X^2 and σ_Y^2 to estimate the standard error of $(\bar{X} - \bar{Y})$.

- ▶ If the **unknown** variances are assumed to be **different**, the standard error of $(\bar{X} - \bar{Y})$ is estimated by replacing σ_X^2 and σ_Y^2 with the sample variances obtained from the two samples, S_X^2 and S_Y^2 .

The confidence intervals built (under appropriate assumptions) in the two cases have different margins of error, due both (obviously) to different standard error estimates and to different reliability factors (percentiles).

CI for difference between means, independent samples

If X and Y have **normal** distributions, and their variances are assumed to be **equal**:

$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{s_{pooled}^2}{n_X} + \frac{s_{pooled}^2}{n_Y}}}$ has a Student's t distribution with $(n_X + n_Y - 2)$ degrees of freedom.

Thus, using the usual procedure, the confidence interval (estimate) is:

$$\begin{aligned} ci_{1-\alpha}(\mu_X - \mu_Y) &= \left[(\bar{x} - \bar{y}) \mp t_{n_X+n_Y-2, \alpha/2} \sqrt{\frac{s_{pooled}^2}{n_X} + \frac{s_{pooled}^2}{n_Y}} \right] \\ &= \left[(\bar{x} - \bar{y}) \mp t_{n_X+n_Y-2, \alpha/2} \cdot se(\bar{X} - \bar{Y}) \right] \end{aligned}$$

It can also be used without assuming normality if the samples are sufficiently large (same considerations on normal vs. Student's t as for interval for non-normal population with unknown variance)!

If X and Y have **normal** distributions (or the samples **are** sufficiently large) with **different** variances, the interval is built using the estimates s_X^2 and s_Y^2 ; in this case, the reliability factor is still the percentile of a Student's t , but with a different number of degrees of freedom (the interval in this case will only be obtained with R^*)

Rstudio: CI.diffmean() for independent samples

The function `CI.diffmean()` in package `VBStats` allows to determine the confidence interval for the difference between means. In the case of **independent** samples, intervals can be built in two alternative ways

Samples contained in two separate vectors

```
CI.diffmean(x, y, type="independent", sigma.x=NULL, sigma.y=NULL,  
            conf.level=0.95, digits=2, data)
```

where:

- ▶ `x, y`: name of two **numeric** vectors possibly names of two variables in `data`. The function determines the CI for $\mu_x - \mu_y$
- ▶ `type="independent"` : specifies that the samples are independent (default) and not paired (case where `type="paired"` must be indicated, as we shall see later)
- ▶ `sigma.x, sigma.y`: allow standard deviations to be specified, if **known**; in the case of known and equal variances, it is sufficient to specify only one of the two arguments
- ▶ `conf_level` is the confidence level (0.95 by default)
- ▶ `digits` is the number of decimals used to round off the reported statistics

Note: the function does not check whether the conditions necessary for the determination of the interval (e.g. sample size, type of samples, paired or independent) are fulfilled

Rstudio: CI.diffmean() for independent samples

The function `CI.diffmean()` in package `UBStats` allows to determine the confidence interval for the difference between means. In the case of **independent** samples, intervals can be built in two alternative ways

Samples obtained based on a 2-level vector

```
CI.diffmean(x, by, type="independent", sigma.by=NULL,  
            conf.level=0.95, digits=2, data)
```

where:

- ▶ **x, by**: vectors of **equal length** ordered in such a way that their elements refer to the same unit; **x** is the name of a **numeric** vector **while** **by** is the name of a vector **that takes only two values**, **by1 < by2** (considering standard, alphabetical, numeric or order of levels in a factor); possibly **x** and/or **by** can be the names of variables in **data**. The function determines the CI for **$\mu_{by1} - \mu_{by2}$**
- ▶ **type="independent"**: specifies that samples are independent (default)
- ▶ **sigma.by**: vector containing the standard deviations **x** in the two groups [see manual*].
- ▶ **conf_level** is the confidence level (0.95 by default)
- ▶ **digits** is the number of decimals used to round off the reported statistics

Note: the function does not check whether the conditions necessary for determining the interval (e.g. sample width, type of samples - paired or independent) are met

* In the case of known variances we suggest using the **x, y** approach

Hands on: difference between means, independent samples

The dataframe `Insurance` (in `Lesson 14-16_Data.Rdata`) contains information on reimbursements (`refunds`, in dollars) for healthcare costs to subscribers (unrelated) with different socio-demographic characteristics (`age`, `sex`, `bmi`, `children`, `smoker`, `region`).

Estimate the difference in the average amount of expenditure reimbursed to smokers and non-smokers using a 99% confidence interval, assuming that the variances are equal

The two samples are **independent** as the subscribers have no links one to each other

```
# approach based on x,y
# extraction of data of interest
refund.smoke <- Insurance$refund[Insurance$smoker=="yes"]
refund.nosmoke<-Insurance$refund[Insurance$smoker=="no"] .

# no need to specify type, as independent=default
CI.diffmean(x=refund.smoke,y=refund.nosmoke,conf.level=0.99,digits=1)

# approach based on x,by
CI.diffmean(x=refund,by=smoker, conf.level=0.99,digits=1,data=Insurance)
```

Hands on: difference between means, independent samples

```
> # Approach based on x,y
> CI.diffmean(x=refund.smoke,y=refund.nosmoke,conf.level=0.99,digits=1)
```

Confidence interval for $\mu_x - \mu_y$

Samples: independent

Confidence level: 0.99

Variances: unknown

x = refund.smoke

y = refund.nosmoke

Unknown variances assumed to be equal

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	274	1064	32050.2	8434.3	23616	11541.5	5993.8	506.1	22312.4	24919.5
Student-t	274	1064	32050.2	8434.3	23616	11541.5	5993.8	506.1	22310.5	24921.4

Unknown variances assumed to be different

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	274	1064	32050.2	8434.3	23616	11541.5	5993.8	721.1	21758.6	25473.3
Student-t	274	1064	32050.2	8434.3	23616	11541.5	5993.8	721.1	21747.2	25484.7

The results show that with 99% confidence the average reimbursement to smokers is greater, as all confidence interval values are positive, regardless of assumptions and approximations, and the mean difference is estimated to be between 22310 and 24921 (if variances are equal)

The function reports the intervals built both assuming that the unknown variances are equal and that they are different. We note different standard error estimates in the two cases and a slightly wider interval in the case of unknown variances. Under both assumptions, both an interval obtained using the normal approximation and an interval obtained using Student's t are reported, which are practically identical, as the percentiles of the two distributions are almost identical given the high sample sizes.

Hands on: difference between means, independent samples

```
> CI.diffmean(x=refund.smoke,y=refund.nosmoke,conf.level=0.99,digits=1)
```

Confidence interval for $\mu_x - \mu_y$

Samples: independent

Confidence level: 0.99

Variances: unknown

x = refund.smoke

y = refund.nosmoke

Unknown variances assumed to be equal

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	274	1064	32050.2	8434.3	23616	11541.5	5993.8	506.1	22312.4	24919.5
Student-t	274	1064	32050.2	8434.3	23616	11541.5	5993.8	506.1	22310.5	24921.4

Comparison of syntax based on **x, y** and syntax based on **x, by** *(output selected: attention limited to case of equal variances)*

```
> # approach based on x,by
```

```
> CI.diffmean(x=refund,by=smoker,conf.level=0.99,digits=1,data=Insurance)
```

Confidence interval for $\mu_x - \mu_y$

Samples: independent

Confidence level: 0.99

Variances: unknown

x = refund|smoker=no

y = refund|smoker=yes

Unknown variances assumed to be equal

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	1064	274	8434.3	32050.2	-23616	5993.8	11541.5	506.1	-24919.5	-22312.4
Student-t	1064	274	8434.3	32050.2	-23616	5993.8	11541.5	506.1	-24921.4	-22310.5

The results are the same regardless of the syntax, but the signs of the intervals are opposite because in the second case we cannot control the order (unless we create a factor by reordering the levels)

Hands on: difference between means, independent samples

The dataframe `Insurance` (in `Lesson 14-16_Data.Rdata`) contains information on reimbursements (`refunds`, in dollars) for healthcare costs to subscribers (unrelated) with different socio-demographic characteristics (`age`, `sex`, `bmi`, `children`, `smoker`, `region`).

Estimate the difference in the average amount of expenditure reimbursed to smokers and non-smokers *without children* using a 99% confidence interval

The two samples are **independent** as the subscribers are not related to each other

```
# extraction of data of interest
refund.smoke <- Insurance$refund[Insurance$smoker=="yes" &
                                Insurance$children==0]
refund.nosmoke<-Insurance$refund[Insurance$smoker=="no" &
                                Insurance$children==0]

CI.diffmean(x=refund.smoke,y=refund.nosmoke,
            type="independent",conf.level=0.99)
```

Hands on: difference between means, independent samples

```
> CI.diffmean(x=refund.smoke,y=refund.nosmoke,  
+            type="independent",conf.level=0.99)
```

Confidence interval for $\mu_x - \mu_y$

Samples: independent

Confidence level: 0.99

Variances: unknown

x = refund.smoke

y = refund.nosmoke

The results show that with 99% confidence the average reimbursement to smokers is also higher in the sub-population of childless subjects, as all values of the confidence interval are positive, regardless of assumptions and approximations

Unknown variances assumed to be equal

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	115	459	31341.4	7611.8	23729.6	11596.7	5858.5	768.3	21750.5	25708.6
Student-t	115	459	31341.4	7611.8	23729.6	11596.7	5858.5	768.3	21743.9	25715.2

Unknown variances assumed to be different

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	115	459	31341.4	7611.8	23729.6	11596.7	5858.5	1115.4	20856.4	26602.7
Student-t	115	459	31341.4	7611.8	23729.6	11596.7	5858.5	1115.4	20813.3	26645.9

The standard errors calculated assuming equal or different unknown variances differ from each other; the differences between the intervals are not marked, even if the interval calculated assuming unknown variances is slightly wider. Given the sample sizes, the percentiles of the standard normal and of the Student's t distribution - and consequently the obtained intervals - are substantially the same

Hands on: difference between means, independent samples

Consider data on the calls to a call centre in the `CallCenter.P` dataframe (in **Lesson 14-16_Data.Rdata**) with particular reference to the reason for the contact (`Topic`) and the duration in seconds of the conversation (`TimeTalking`)

Estimate with a 95% confidence interval the difference between the average talk time of contract-related and payment-related calls (`Topic="Contract related"` and `Topic="Payment related"`), under the assumption that the variances are equal, and assuming first that they are known and equal to 22500 seconds² and then that they are unknown

```
# Extract the data of interest
talk.contract<-CallCenter.P$TimeTalk[CallCenter.P$Topic=="Contract related"].
talk.payment<-CallCenter.P$TimeTalk[CallCenter.P$Topic=="Payment related"].
CI.diffmean(x=talk.contract,y=talk.payment,sigma.x=sqrt(22500),
            type="independent",conf.level=0.95,digits=1)
CI.diffmean(x=talk.contract,y=talk.payment,type="independent",
            conf.level= 0.95,digits=1)
```

Hands on: difference between means, independent samples

95% confidence interval the difference between the average talk time of contract-related and payment-related calls (Topic="Contract related" and "Payment related"), under the assumption that the variances are equal, assuming first that they are known and equal to 22500 seconds² and then that they are unknown

```
Confidence interval for  $\mu_x - \mu_y$ 
```

```
Samples: independent
```

```
Confidence level: 0.95
```

```
Variances: known
```

```
  x = talk.contract
```

```
  y = talk.payment
```

n_x	n_y	xbar	ybar	xbar-ybar	sigma_X	sigma_Y	SE	Lower	Upper
82	80	233.1	257	-23.9	150	150	23.6	-70.1	22.3

```
Confidence interval for  $\mu_x - \mu_y$ 
```

```
Samples: independent
```

```
Confidence level: 0.95
```

```
Variances: unknown
```

```
  x = talk.contract
```

```
  y = talk.payment
```

The interval built without assuming that the variances are known is wider. However, in both cases we conclude (with 95% confidence) that the difference between the means can be positive or negative, despite a negative point estimate, regardless of the assumptions on the variances

```
Unknown variances assumed to be equal
```

	n_x	n_y	xbar	ybar	xbar-ybar	s_X	s_Y	se	Lower	Upper
Normal.Approx	82	80	233.1	257	-23.9	175.4	152.9	25.9	-74.6	26.8
Student-t	82	80	233.1	257	-23.9	175.4	152.9	25.9	-75.0	27.2

CI for difference between means, paired samples

In the case of **paired samples** of size n , two measurements X_i and Y_i are available for each sample unit; the differences $D_i = X_i - Y_i$ can therefore be considered as measurements on the r.v. $D = X - Y$ on n sample units.

In this case, the variance of the sample mean of the differences, $\bar{D} = \bar{X} - \bar{Y}$ is:

$$\begin{aligned} \text{Var}(\bar{D}) &= \text{Var}(D)/n = \sigma_D^2/n \\ &= \text{Var}(X - Y)/n = [\text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)]/n = [\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}]/n \end{aligned}$$

Assuming that X e Y are **jointly normal** (or that the **sample size is sufficiently large**), the statistic $\bar{D}$ has a **normal** distribution **and** the confidence interval for $\mu_X - \mu_Y$ is

$$ci_{1-\alpha}(\mu_X - \mu_Y) = [\bar{d} \mp z_{\alpha/2} \cdot \sigma_D/\sqrt{n}] = [\bar{d} \mp z_{\alpha/2} \cdot SE(\bar{D})]$$

where $\bar{d} = \bar{x} - \bar{y}$ is the sample mean of the realisations of the differences $d_1, \dots, d_n$

CI for difference between means, paired samples

The variance of the differences (and/or the variances of the two populations and their covariance) is seldom known, and the variance of $\bar{D} = \bar{X} - \bar{Y}$ must be estimated - as we have seen when dealing with point estimation - with the **corrected sample variance**:

$$s_D^2 = \sum_{i=1}^n \frac{(D_i - \bar{D})^2}{n-1} = s_X^2 + s_Y^2 - 2s_{XY}$$

Since $\bar{D}$ is a sample mean, in the case of a **normal** distribution, the confidence interval for $(\mu_X - \mu_Y)$ is obtained exactly as the interval for the mean of a normal population with unknown variance:

$$ci_{1-\alpha}(\mu_X - \mu_Y) = [\bar{d} \mp t_{n-1, \alpha/2} \cdot s_D / \sqrt{n}] = [\bar{d} \mp t_{n-1, \alpha/2} \cdot se(\bar{D})]$$

If the population is not normal, but the sample is large enough, so that the central limit theorem applies, it is possible to use the interval based on the percentile of the standard normal or on the percentile of the Student's distribution, in order to account for the uncertainty associated with the estimate of σ_D^2

Rstudio: CI.diffmean() for paired samples

The function `CI.diffmean()` in package `VBStats` allows to determine the confidence interval for the difference between means of **paired** samples, with the following syntax:

```
CI.diffmean(x, y, type="paired", sigma.d=NULL, conf.level=0.95,  
            digits=2, date)
```

where:

- ▶ `x, y`: name of two **numeric** vectors ordered so that their elements refer to the same unit or names of two numeric columns in the dataframe specified in `date`
- ▶ `type="paired"` : specifies that the samples are paired
- ▶ `sigma.d`: allows the standard deviation of the **difference** to be specified, if **known**
- ▶ `conf_level` is the confidence level (0.95 by default)
- ▶ `digits` is the number of decimals used to round off the reported statistics

Remark: the function does not check whether the conditions necessary for determining the interval (e.g. sample width, type of samples - paired or independent) are met

Hands on: difference between means, paired samples

A clinical study conducted on 44 patients with high (bad) cholesterol levels to evaluate the efficacy of a cholesterol-lowering drug reported the following results:

Cholesterol level at the beginning of treatment: $\bar{x}_{ini} = 155$; $s_{ini} = 19.9$

Cholesterol level at 8 weeks: $\bar{x}_{8w} = 122$; $s_{8w} = 24.5$, $s_{ini,8w} = 23.4$

sample covariance between x_{ini} , x_{8w}

Estimate the average reduction in cholesterol level following treatment using a 90% confidence interval

The sample is not particularly large, although the distribution of the sample mean can be approximated by a normal one already for samples of size 30

```
> xbar_ini <- 155
> s_ini <- 19.9
> xbar_8w <- 122
> s_8w <- 24.5
> cov_ini_8w <- 23.4
> n <- 44
> mean_ini_8w <- xbar_ini - xbar_8w # difference between averages
> se_diff <- sqrt(s_ini^2 + s_8w^2 - 2*cov_ini_8w) / sqrt(n)
> t_43_alpha_2 <- qt(0.95, df=43)
> ME <- t_43_alpha_2 * se_diff
> c((mean_ini_8w - ME), (mean_ini_8w + ME))
[1] 25.19095 40.80905
```

Hands on: difference between means, paired samples

The dataframe `Fatalities_US` (in `Lesson 13_Data.Rdata`) contains data on the number of **Accidents** (per 100,000 inhabitants) registered in 48 US states. General information on states is available (unemployment rate, % of drivers aged between 15 and 24, `YoungDrivers`), besides the number of accidents at night (`Night`) and due to alcohol (`Alcohol`). To fight alcohol-related accidents, an awareness-raising campaign was carried out, following which a new survey of the number of such accidents (per 100,000 inhabitants, `AlcoholPostCampaign`) was collected.

Evaluate the reduction in the average number of alcohol-related accidents as a result of the campaign using a 95% confidence interval.

```
> CI.diffmean(x=Alcohol,y=AlcoholPostCampaign, type="paired",  
+             conf.level=0.95,data=Fatalities_US)
```

Confidence interval for $\mu_x - \mu_y$

Samples: paired

Confidence level: 0.95

Variance: unknown

x = Alcohol

y = AlcoholPostCampaign

Based on these results, we can conclude (with 95% confidence) that the average number of alcohol-related accidents decreased after the campaign

	n	xbar	ybar	d=xbar-ybar	s_D	se	Lower	Upper
Normal.Approx	48	7.47	6.18	1.28	2.31	0.33	0.63	1.93
Student-t	48	7.47	6.18	1.28	2.31	0.33	0.61	1.95

Interval estimation

Confidence intervals for the difference between proportions

Confidence interval for the difference between proportions

As already seen with reference to point estimation, in many applications we are interested in comparing the proportions of a certain phenomenon in two populations

Population 1	Population 2
$X_1, \dots, X_{n_X}$ sample of size n_X , iid X distributed according to a Bernoulli of parameter p_X (i.e. X takes on a value of 1 or 0 depending on whether a success is observed or not, and p_X = proportion of successes in the population), with $E(X) = p_X$ and $Var(X) = p_X(1 - p_X)$	$Y_1, \dots, Y_{n_Y}$ sample of size n_Y iid Y distributed according to a Bernoulli parameter p_Y (i.e. Y takes on a value of 1 or 0 depending on whether a success is observed or not, and p_Y = proportion of successes in the population), with $E(Y) = p_Y$ and $Var(Y) = p_Y(1 - p_Y)$
$\hat{P}_X = (X_1 + \dots + X_{n_X})/n_X$ = sample proportion of successes	$\hat{P}_Y = (Y_1 + \dots + Y_{n_Y})/n_Y$ = sample proportion of successes

The parameter of interest is the difference between the proportions in the two populations,
 $p_X - p_Y$

Confidence interval for the difference between proportions

The point estimator for the difference between the proportions in two populations is the difference between the proportions in the two samples, $\hat{P}_X - \hat{P}_Y$

In the case of **independent samples**, we have seen that:

$$Var(\hat{P}_X - \hat{P}_Y) = \frac{p_X(1 - p_X)}{n_X} + \frac{p_Y(1 - p_Y)}{n_Y}$$

$Var(\hat{P}_X - \hat{P}_Y)$ is **never** known (it depends on p_X and p_Y), but it can be estimated by replacing p_X and p_Y with the observed sample proportions.

If the samples are large enough, by the central limit theorem we have:

$$\begin{aligned} ci_{1-\alpha}(p_X - p_Y) &= \left[(\hat{p}_X - \hat{p}_Y) \mp z_{\alpha/2} \sqrt{\frac{\hat{p}_X(1 - \hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1 - \hat{p}_Y)}{n_Y}} \right] = \\ &= \left[(\hat{p}_X - \hat{p}_Y) \mp z_{\alpha/2} se(\hat{P}_X - \hat{P}_Y) \right] \end{aligned}$$

Note that the percentile of the standardised normal distribution is used here.

Rstudio: CI.diffprop() for independent samples

The function `CI.diffprop()` in package `UBStats` allows to determine the confidence interval for the difference between the proportions of 'successes' in two independent populations. Again, it is possible to use the syntax based on two vectors `x, y` or the syntax based on **a vector `x`** partitioned on the **basis of a 2-level vector `by`**. In the **first case**:

```
CI.diffprop(x, y, success.x=NULL, success.y=NULL,  
            conf.level=0.95, digits=2, data)
```

where:

- ▶ `x, y`: name of two vectors or factors or two variables in `data`: the interval is built for **`p_x - p_y`**
- ▶ `success.x, success.y`: define the values of `x` and `y` to be taken as success in the two samples. They can be omitted (in one or both cases) if `x` or `y` are binary (values 0/1; in this case it is assumed that 1 indicates a success) or logical (values **`TRUE/FALSE`**; in this case it is assumed that **`TRUE`** indicates a success). If success is encoded in the same way in the two samples, it is sufficient to specify only one of the two arguments
- ▶ `conf_level` is the confidence level (0.95 by default)
- ▶ `digits` is the number of decimals used to round off the reported statistics

Note that the function does not check whether the conditions necessary for determining the interval (e.g. sample width) are satisfied

Rstudio: CI.diffprop() for independent samples

The function `CI.diffprop()` in package `UBStats` allows to determine the confidence interval for the difference between the proportions of 'successes' in two independent populations.

In the case of vector `x` partitioned on the basis of a 2-level vector `by`:

```
CI.diffprop(x, by, success.x=NULL, conf.level=0.95, digits=2, data)
```

where:

- ▶ `x, by`: vectors (or factors) with the same length ordered so that their elements refer to the same unit. `x` is the name of a vector/factor, `by` is the name of a two-level vector/factor with two values/levels `by1 < by2` (possibly `x` and/or `by` can be the names of variables in `data`). In this case, the function calculates the interval for `p_by1 - p_by2`
- ▶ `success.x`: defines the values of `x` to be taken as success. It can be omitted when `x` is binary (values 0/1; in this case, 1 is assumed to indicate success) or logical (values `TRUE/FALSE`; in this case, `TRUE` is assumed to indicate success).
- ▶ `conf_level` is the confidence level (0.95 by default)
- ▶ `digits` is the number of decimals used to round off the reported statistics

Note that the function does not check whether the conditions necessary for determining the interval (e.g. sample width) are satisfied

Hands on: difference between proportions

The dataframe **Banner** (in **Lesson 14-16_Data.Rdata**) contains information on two alternative banners (A and B) shown to users of a social network. It is of interest to compare the Click-Through Rate (CTR) of the two banners, i.e. the ratio between the number of clicks on each banner and the number of banner views served by the ad server (impressions). The variable **Click** indicates for each user who viewed a banner whether they clicked (**Yes**) or not (**No**).

Build a 90% confidence interval for the difference between the CTRs

```
# syntax x,y
# define data vectors
banner_A<-Banner$Click[Banner$Banner=="A"]
banner_B<-Banner$Click[Banner$Banner=="B"]

CI.diffprop(x=banner_A,y=banner_B,success.x="Yes",
            conf.level=0.90,digits=4)

# syntax x,by
CI.diffprop(x=Click,y=Banner,success.x="Yes",conf.level=0.90,
            digits=4,data=Banner)
```


Hands on: difference between proportions

```
> CI.diffprop(x=banner_A,y=banner_B,success.x="Yes",
+             conf.level=0.90,digits=4)
```

Warning:

Only 'success.x' is specified; 'success.y' set equal to 'success.x'.

Confidence interval for $p_x - p_y$

Confidence level: 0.9

x:banner_A = Yes

y:banner_B = Yes

n _x	n _y	phat _x	phat _y	phat _x -phat _y	s _X	s _Y	se	Lower	Upper
2364	2323	0.1929	0.263	-0.0701	0.3946	0.4403	0.0122	-0.0902	-0.05

```
>
```

```
> CI.diffprop(x=Click,by=Banner,success.x="Yes",
+             conf.level=0.90,digits=4,data=Banner)
```

Confidence interval for $p_x - p_y$

Confidence level: 0.9

x:Click = Yes | Banner = A

y:Click = Yes | Banner = B

n _x	n _y	phat _x	phat _y	phat _x -phat _y	s _X	s _Y	se	Lower	Upper
2364	2323	0.1929	0.263	-0.0701	0.3946	0.4403	0.0122	-0.0902	-0.05

What considerations?

Recap: averages and differences between averages

Parameter	Estimator	Estimate	Samples and Variance(s)		Standard error, <i>SE</i> or <i>se</i>	Pop. Distr	Percentile
μ	$\bar{X}$	$\bar{x}$	σ known		$SE = \sigma/\sqrt{n}$	Normal/large samples	Normal
			σ unknown		$se = s/\sqrt{n}$	Normal/large samples	Student's t^* ($n-1$)
$\mu_X - \mu_Y$	$\bar{X} - \bar{Y}$	$\bar{x} - \bar{y}$	Independent Samples	σ_X^2 and σ_Y^2 known	$SE = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$	Both Normal or Large samples	Normal
				$\sigma_X^2 = \sigma_Y^2$ unknown	$se = \sqrt{\frac{s_{Pool}^2}{n_X} + \frac{s_{Pool}^2}{n_Y}}$	Both Normal or Large samples	Student's t^* (n_X+n_Y-2)
				$\sigma_X^2 \neq \sigma_Y^2$ unknown	$se = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$	Both Normal or Large samples	Student's t (only with R)
			Paired Samples	σ_D^2 (or $\sigma_X^2, \sigma_Y^2, \sigma_{XY}$) known	$SE = \sqrt{\frac{\sigma_D^2}{n}} = \sqrt{\frac{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}{n}}$	Joint Normal or Large samples	Normal
				σ_D^2 (or $\sigma_X^2, \sigma_Y^2, \sigma_{XY}$) unknown	$se = \sqrt{\frac{s_D^2}{n}} = \sqrt{\frac{s_X^2 + s_Y^2 - 2s_{XY}}{n}}$	Joint Normal or Large samples	Student's t^* ($n-1$)

* In the case of unknown variance(s) for large samples the approximation is **normal**, but one can use Student's t to widen the interval and thus account for the uncertainty associated with estimating the variance(s)

Recap: proportion and difference between proportions

Parameter	Estimator	Estimate		Samples	Standard error, <i>SE</i> or <i>se</i>	Pop. Distr	Percentile
p	$\hat{P}$	$\hat{p}$			$se = \sqrt{\hat{p}(1 - \hat{p})/n}$	Large sample	Normal
$p_X - p_Y$	$\hat{P}_X - \hat{P}_Y$	$\hat{p}_X - \hat{p}_Y$		Independent	$se = \sqrt{\frac{\hat{p}_X(1 - \hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1 - \hat{p}_Y)}{n_Y}}$		